# Comparing models with ANOVA

Each part has two scenarios with the same tasks. We'll work through scenario **A** together as a class. Then you'll work through scenario **B** in your group.

Every question in this handout compares a **simpler** model to a more **complex** model. Refer back to this box as you go.

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$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \;/\; (p_{complex} - p_{simple})}{\text{RSS}_{complex} \;/\; (n - p_{complex})}$$

- RSS: residual sum of squares, $\sum(y_i - \hat{y}_i)^2$. The variation the model leaves unexplained.
- $p$: number of coefficients the model estimates, including the intercept
- $n$: number of observations
- $df = n - p$: degrees of freedom left over after fitting the model
- **Numerator**: improvement in fit per df spent. **Denominator**: leftover noise per df remaining.

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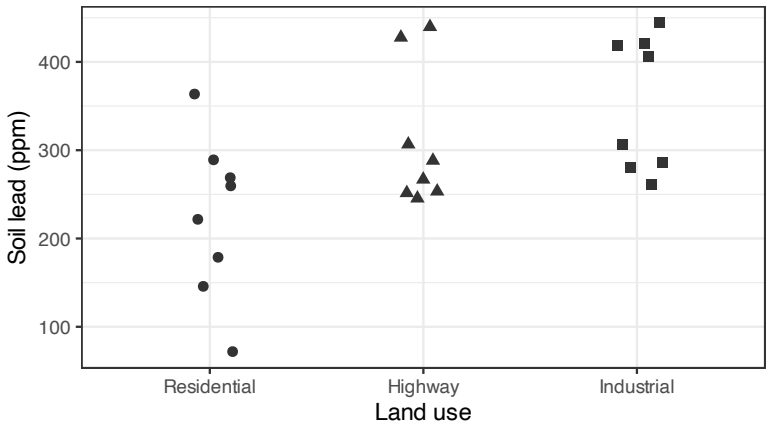
## Part 1: Three levels

For each scenario:

1. **Dummy variables.** Fill in the dummy variable table.
2. **Full model.** Write the equation for the full model.
3. **Draw the means.** Use the coefficients to draw the intercept-only model and each category's mean on the figure.
4. **Degrees of freedom and F.** Fill in $p$ and $df$ for each model, then calculate $F$.
5. **Interpretation.** State $H_0$ and $H_A$. Then interpret the p-value for your $F$ in a sentence.

## 1A. Soil lead (together)

Lead from old gasoline and industrial emissions settles into soil, where children can be exposed. A county health department sampled soil **lead** (ppm) in 8 yards near each of three land uses (**land_use**): residential streets, a highway, and an industrial site.



## Dummy variables

| Land use | Highway | Industrial |
|---|---|---|
| Residential | _____ | _____ |
| Highway | _____ | _____ |
| Industrial | _____ | _____ |

## Full model

## Draw the means

```
coef(lm(lead ~ 1, yards))
```

```
(Intercept)
   295.9625
```

```
coef(lm(lead ~ land_use, yards))
```

```
       (Intercept)    land_useHighway land_useIndustrial
          224.9125            85.0625           128.0875
```

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `lead ~ 1` | _____ | _____ | 210,538.5 |
| `lead ~ land_use` | _____ | _____ | 142,556.7 |

## Interpretation

State $H_0$ and $H_A$.

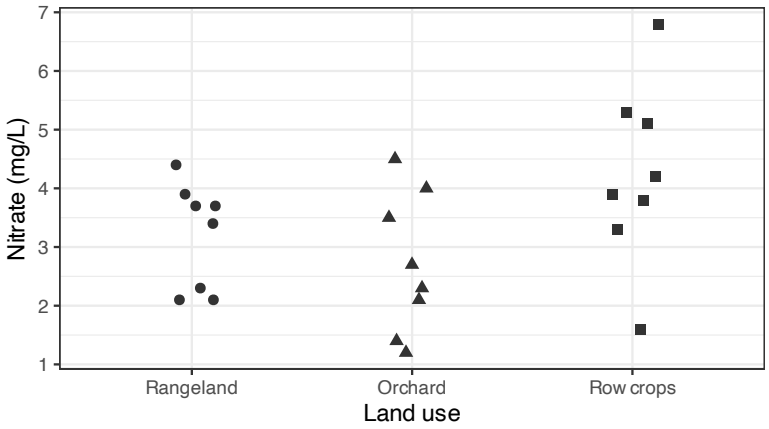
$H_0$:

$H_A$:

The p-value for this $F$ is p = 0.017. Interpret it in a sentence.

# 1B. Groundwater nitrate (in groups)

Fertilizer applied to crops can leach into groundwater as nitrate, contaminating drinking water wells. A water agency measured **nitrate** (mg/L) in 8 wells under each of three land uses (**land_use**): rangeland, orchards, and row crops.




## Dummy variables

| Land use | Orchard | Row crops |
|---|---|---|
| Rangeland | _____ | _____ |
| Orchard | _____ | _____ |
| Row crops | _____ | _____ |

## Full model

## Draw the means

```
coef(lm(nitrate ~ 1, wells))
```

```
(Intercept)
     3.3875
```

```
coef(lm(nitrate ~ land_use, wells))
```

```
      (Intercept)   land_useOrchard land_useRow crops
           3.2000           -0.4875            1.0500
```

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `nitrate ~ 1` | _____ | _____ | 42.2 |
| `nitrate ~ land_use` | _____ | _____ | 32.3 |

## Interpretation

$H_0$:

$H_A$:

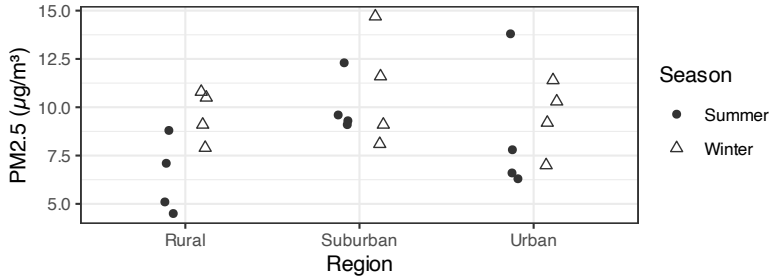
The p-value for this F is p = 0.061. Interpret it in a sentence.

# Part 2: Two categorical predictors

Now there are two categorical predictors. Does the second one justify its cost *after* accounting for the first? Same tasks as Part 1, minus the dummy table.

# 2A. Air quality (together)

Fine particulate matter (PM2.5) damages lungs and hearts, and winter wood smoke can trap it near the ground. An air district measured PM2.5 (**pm**, µg/m³) on 4 days for each combination of **region** (rural, suburban, or urban) and **season** (summer or winter). Does season matter after accounting for region?




## Full model

## Draw the models

```
coef(lm(pm ~ region, air))
```

```
   (Intercept) regionSuburban    regionUrban 
         7.975          2.500          1.075 
```

```
coef(lm(pm ~ region + season, air))
```

```
   (Intercept) regionSuburban    regionUrban   seasonWinter 
      7.166667       2.500000       1.075000       1.616667 
```

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `pm ~ region` | _____ | _____ | 120.4 |
| `pm ~ region + season` | _____ | _____ | 104.7 |

## Interpretation

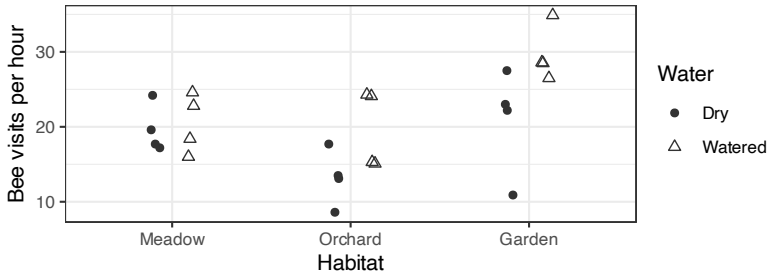
$H_0$:

$H_A$:

The p-value for this F is p = 0.099. Interpret it in a sentence.

# 2B. Bees and watering (in groups)

Watered flowers may produce more nectar, attracting more bees. A pollinator team counted bee **visits** per hour to 4 flower patches for each combination of **habitat** (meadow, orchard, or garden) and **water** (dry or watered). Does watering matter after accounting for habitat?




## Full model

## Draw the models

```
coef(lm(visits ~ habitat, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden 
       20.0625        -3.6000         5.2000 
```

```
coef(lm(visits ~ habitat + water, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden   waterWatered 
         17.400          -3.600           5.200           5.325 
```

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `visits ~ habitat` | _____ | _____ | 626.7 |
| `visits ~ habitat + water` | _____ | _____ | 456.5 |

## Interpretation

$H_0$:

$H_A$:

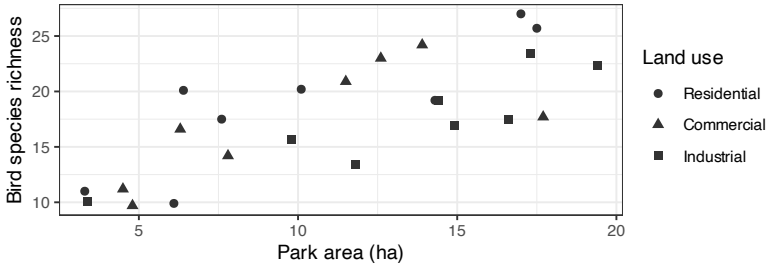
The p-value for this $F$ is p = 0.013. Interpret it in a sentence.

# Part 3: Three levels plus a continuous predictor

Now the second predictor is continuous. Does the categorical predictor justify its cost *after* accounting for the continuous one? Same tasks as Part 2.

# 3A. Birds in city parks (together)

City parks can be refuges for birds, and bigger parks usually hold more species. Does what surrounds a park matter too? A city surveyed bird species **richness** in 8 parks in each of three surrounding land uses (**land_use**): residential, commercial, and industrial. It also recorded each park's **area** (ha).




## Full model

## Draw the models

```
coef(lm(richness ~ area, parks))
```

```
(Intercept)        area
  9.2357591   0.7618654
```

```
coef(lm(richness ~ area + land_use, parks))
```

```
       (Intercept)              area land_useCommercial land_useIndustrial
         9.9183439         0.8657746         -1.2911902         -4.2505121
```

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `richness ~ area` | _____ | _____ | 257.9 |
| `richness ~ area + land_use` | _____ | _____ | 188.4 |

## Interpretation

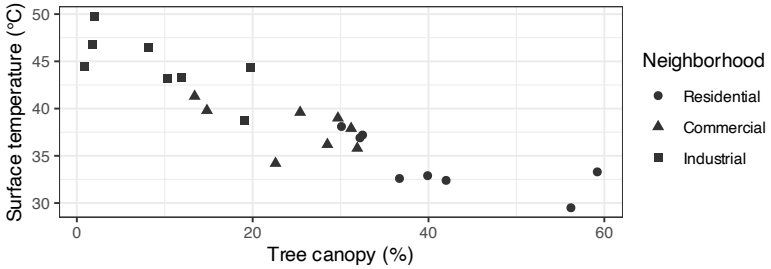
$H_0$:

$H_A$:

The p-value for this F is p = 0.043. Interpret it in a sentence.

# 3B. Urban heat (in groups)

Pavement and rooftops soak up summer sun, while trees shade the ground. A city measured summer surface temperature (**temp**, °C) at 8 sites in each of three **neighborhood** types: residential, commercial, and industrial. It also recorded the tree **canopy** cover (%) around each site. Does neighborhood type matter *after* accounting for canopy?




## Full model

## Draw the models

```
coef(lm(temp ~ canopy, sites))
```

```
(Intercept)      canopy
 46.3901866  -0.2991744
```

```
coef(lm(temp ~ canopy + neighborhood, sites))
```

```
          (Intercept)                canopy neighborhoodCommercial neighborhoodIndustrial
           43.2780196            -0.2230053              0.2024248              3.4194922
```

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `temp ~ canopy` | _____ | _____ | 122.1 |
| `temp ~ canopy + neighborhood` | _____ | _____ | 96.1 |

## Interpretation

$H_0$:

$H_A$:

The p-value for this F is p = 0.091. Interpret it in a sentence.